Canonical KB vs. Swarm Delta vs. Raw Caption — Validation Report

Disease: Soybean :: Southern Blight | Image: bugwood::5581651 | Generated 2026-05-16

1. Disease & canonical KB (Phase-1, on disk)

- **Disease:** Southern Blight | **Pathogen:** Agroathelia rolfsii | **Type:** Fungal
- **Affected parts:** Foliar, Stem, Root, Whole plant | **KB confidence:** high | **# sources:** 2

Canonical `visual_symptoms.summary`:

> Seedling infection causes pre- or post-emergence damping-off; later in the season entire plants yellow and wilt with leaves turning brown and remaining attached, accompanied by a dark brown girdling lesion at the soil surface and white fanlike mycelium with small round sclerotia on the lower stem.

Canonical `diagnostic_features`:

> Conspicuous white, fanlike mats of fungal mycelium at the base of the stem with numerous mustard-seed-sized sclerotia that progress from yellow-tan to reddish-brown to dark brown at maturity.

Canonical source quote (Crop Protection Network):

> "Seedling infection results in pre- or post-emergence damping-off. Later in the season, entire plants may become yellow and wilt, with leaves turning brown and often remaining attached to the plant. A dark brown lesion that girdles the stem occurs at the soil surface. This lesion is generally accompanied by the development of conspicuous white, fanlike mats of fungal mycelium that form on the base of the stem, on leaf residue, and on the soil surface around infected plants. Numerous, small round fungal bodies that are about the size of mustard seeds (called sclerotia) form on these fungal mats and on the lower stem."

2. Per-specialist table — canonical vs. delta vs. raw caption vs. validation

Image bugwood::5581651, all specialist outputs, both rounds. Note how `canonical_says` is "(not specified)" in round 1 even though the canonical KB above clearly contains these facts — the canonical slice is not reaching round-1 agents, so confirmations are mis-logged as net-new deltas.

Specialist (round, conf)	Field	Canonical KB passed to agent (`canonical_says`)	DELTA added by specialist (`image_shows`)	Raw caption (`image_quote`)	Online-source validation
RootAgent (R1, high)	root_visible	(not specified)	The image shows the roots of a plant.	The image displays a close-up view of the roots of a plant.	N/A — organ-presence only, no symptom claim (not validatable)

Specialist (round, conf)	Field	Canonical KB passed to agent (`canonical_says`)	DELTA added by specialist (`image_shows`)	Raw caption (`image_quote`)	Online-source validation
ColorPaletteAgent (R1, high)	color_palette	(not specified)	The dominant color is tan, representing the affected root tissue, with a secondary color of rust-orange representing the sclerotia.	The roots appear tan and covered with a fanlike structure of rust-orange, indicating sclerotia.	SUPPORTED — sclerotia white->tan->reddish-brown is documented [CPN, NCState, Bugwood]
SeverityVisualAgent (R1, high)	severity_visible	(not specified)	The image shows extensive mycelial growth covering much of the stem.	The plant has a tangled mass of white, fan-like mycelium.	SUPPORTED — abundant white mycelium on lower stem documented [APS, MSU]
SporulationAgent (R1, high)	sporulation	(not specified)	White, fanlike mycelium with small round sclerotia on the lower stem.	white fanlike mycelium with small round sclerotia on the lower stem	STRONGLY SUPPORTED — diagnostic hallmark, verbatim match [APS, CPN, Wisconsin]
RootAgent (R2, high)	root_visible	(not specified)	The image displays a plant's root system.	The photo shows a plant's root system with visible roots.	N/A — organ-presence only, no symptom claim (not validatable)
ColorPaletteAgent (R2, high)	color_palette	dark brown girdling lesion at the soil surface	The affected roots show a mix of brown and tan colors.	The roots exhibit a complex pattern of brown and tan hues.	SUPPORTED — dark-brown girdling lesion + white mycelium documented [APS, CPN]
ColorPaletteAgent (R2, high)	color_palette	(not specified)	The mycelium appears white and fanlike with a hint of off-white.	The mycelium is predominantly white with some off-white tones.	SUPPORTED — dark-brown girdling lesion + white mycelium documented [APS, CPN]
SeverityVisualAgent (R2, high)	severity_visible	White, fanlike mycelium with small round sclerotia on the lower stem.	The image shows extensive white, fanlike mycelium with numerous small round sclerotia on the lower stem.	The mycelium covers a large portion of the visible root structure.	STRONGLY SUPPORTED — textbook hallmark, exact match [APS, CPN, NCState]
LookAlikeCoTAgent (R2, high)	look_alikes_visual	(not specified)	The photo shows extensive mycelial growth covering much of the stem.	The image shows extensive mycelial growth covering much of the stem.	WEAK — generic restatement, no look-alike differential offered
SporulationAgent (R2, high)	sporulation	(not specified)	Mycelium is visible on the roots, extending upwards towards the stem.	The image shows extensive mycelial growth covering much of the stem.	SUPPORTED — mycelium on lower stem/soil/roots documented [APS, CPN]

3. Online validation — method & verdict

Method. Each specialist claim was checked against independent plant-pathology authorities (not just the single source the canonical cites). Queried sources:

- APSnet — *Southern blight, southern stem blight, white mold*
- Crop Protection Network — *Southern Blight of Soybeans*
- NC State Extension — *Southern Blight of vegetable crops*

- University of Wisconsin Horticulture Extension — *Southern Blight*
- Michigan State University IPM — *Southern blight*
- Bugwoodwiki — *Sclerotium rolfsii*

Consensus ground truth. White, fan-shaped mycelial mat at the lower stem / soil line; numerous small (0.5–2 mm, mustard-seed-sized) round sclerotia progressing white -> tan -> reddish-brown -> dark brown; dark brown girdling lesion at the soil surface.

Verdict.

- **SporulationAgent** and **SeverityVisualAgent** reproduce the diagnostic hallmark *verbatim* — STRONGLY SUPPORTED. Core signal is accurate and trustworthy.
- **ColorPaletteAgent** is SUPPORTED on this image (tan/brown + white mycelium), though on other images of this profile it emits an unsupported "olive-green" reading (background-foliage contamination).
- **RootAgent** entries are tautological organ-presence statements with no symptom content — not validatable, low KB value.
- **LookAlikeCoTAgent** restates the dominant sign without offering any differential — WEAK.

Bottom line. The model's *perception* is accurate (5 of 5 substantive claims supported by independent sources), but (a) the canonical slice is not grounding round-1 specialists, so confirmations are mis-logged as deltas, and (b) one true fact is echoed by ~5 off-lane agents, inflating the KB and driving the consolidator JSON past its token budget.